

Examiner reports

Q1.

There were some good attempts at parts (a) and (b). In (a) students were able to use the binomial distribution function on their calculators to obtain the correct final answer. In (b) students generally recognised the need to cube their answer to (a), however it was disappointing that some multiplied their answer to (a) by 3, obtaining an answer greater than 1.

In part (c) many students did not state their assumptions in context or instead gave 'scientific/practical' rather than 'statistical' assumptions. Examples include: all extraneous variables kept constant, the same darts and dart board are used and the distance from the dart board remains the same each time. It is important that students know the correct distributional assumptions for the binomial distribution and are able to apply them to the context of a question. In this case a 'fixed number of trials' was not given credit as this was stated in the question.

Q2.

This was a very well answered question overall. Parts (a), (b), (c) and (d) were all answered correctly by the majority of students, with many making very good use of the calculator. Only part (e) proved problematic, with many students failing to score, often as a result of making comments not clearly enough related to the binomial conditions, or otherwise by not stating them sufficiently in context.

Q7.

This was another well-answered question with the use of tables resulting in full or almost full marks. Answers to part (a) were almost always correct. Students sometimes lost marks in part (b) through the use of $1 - 0.8849$ or by simply stating 0.8074 or 0.8849. Most students also scored marks in part (c) but 'at least' and 'at most' lead to the usual confusions; here between 0.9852 and 0.9680 and / or between 0.0086 and 0.0238.

Q8.

Not surprisingly, given a similar question on at least one previous paper, most students produced very sound, if not fully correct, answers in part (a) by standardising correctly in (i) and then identifying the correct related area in (ii). Some students were caught out in (iii) by quoting zero rather than unity. In part (b), most students used Table 4 correctly to obtain $z = 1.7507$ and then solve the simple equation for x . Correct answers to part (c)(i) proved to be beyond most students. Whilst most realised that their answer to part (a)(ii) was needed for p , almost all ignored 'at least 5' and so gave an answer of p^6 .

Q9.

Almost all students scored the mark in part (a)(i) with slightly fewer scoring the 2 marks in part (a)(ii) where only 1 mark was available for $1 - 0.9597$. It was quite rare to see an incorrect answer to part (a)(iii) where an evaluation of a binomial term was required. A small minority failed to evaluate the correct expression correctly (loss of 1 mark) or attempted to use Table 1 and averaging values for 0.15 and 0.20 (loss of both marks). In part (a)(iv), despite the example given in the question, about 50% of students scored no marks due to not finding the correct value of 0.35 for p . The common error was $p = 0.85 \times$

0.5 = 0.425 followed by interpolation from Table 1 or use of a calculator's in-built function. Some students even tried to evaluate the probability for $p = 0.85$ and $p = 0.85$ separately and then subtract the two resulting probabilities. Similarly, in part(b), many students failed to find the correct value of p with 0.85×0.175 as a common incorrect attempt. A minority of students scored 1 mark for 0.325 or 13.

Q10.

This question proved a good source of marks for almost all candidates; clearly they had a very sound grasp of the applications of binomial distributions. By far the most common approach in part (a) was to use the formula in part (a)(i) and then tables in parts (a)(ii) and (a)(iii). In part (a)(iii), the usual confusions sometimes arose as to whether to use the values for $x = 9$ or 10 or for $x = 4, 5$ or 6. In part (b), the only noticeable error was to evaluate $(32 \times 0.15 \times 0.85)$ as 4.08 but then not take the square root; a loss of 1 mark here. Almost all candidates calculated correctly, using their calculators' mean and standard deviation functions, the mean and the standard deviation of the given 13 values. Many then went on to compare the 2 mean values and the 2 standard deviation values and so comment on the proposed model's suitability. Centres are advised that comments of the form 'means and standard deviations are different' is not sufficiently clear for the awarding of marks.

Q11.

There were many good answers scoring most, sometimes all, of the marks on this binomial question. In part (a), almost all students were able to write down the correct formula and then evaluate correctly $P(X = 2 \mid B(10, 0.275))$. In part (b), it was also pleasing to see the high proportion of students who scored the 5 marks by using the tables correctly for $B(40, 0.35)$ but, as is the norm, some students have yet to master the correct table readings for 'no more than', 'at least' and 'at most'. Sadly, some students equated $(1 - 0.65)$ to 0.45 which proved somewhat disastrous. In order to use tables, part (c) required changing from $B(40, 0.85)$ to $B(40, 0.15)$; a non-too-easy process. Hence, it was most gratifying to see a number of very sound, even fully correct, solutions using tables. A minority of students used $B(40, 0.85)$ and their calculators' in-built cumulative binomial function with similar levels of success.

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Q12.

Stronger students often scored full marks on this question, perhaps due to the question not requiring a qualitative response. Most students scored well in part (a) as they were usually well aware of how to standardise or how to use their calculators' in-built function. When errors did occur it was often down to rounding z -values to one decimal place thereby obtaining inaccurate answers or, particularly in part (a)(ii), finding the complement of the probability required. In part (b)(i), most students obtained 0.372 although some methods were not always clear and others were somewhat contrived. Notable among these was $[0.59871 + 0.77337 - 1]$ which could be argued to be less convincing than $[0.59871 + (1 - 0.77337)]$, particularly in a 'show that' request? Nevertheless, this and other unusual approaches that were justifiable were awarded the 2 marks available. This was not the case for those students who relied entirely on their calculators' in-built function to such an extent that they simply stated $[(P(X < 5.3) - P(X < 5.1)) \text{ or } (P(Z < 0.25) - P(Z < -0.75))] = 0.372 \dots = 0.372$. Although questions similar to part (b)(ii) have appeared fairly regularly on previous papers, many students still struggle to

distinguish between the two situations. Typical approaches were to use $\frac{0.2}{\sqrt{4}}$ in part (b)(ii) and then use $\frac{0.5}{\sqrt{6}}$ (right!). Other less common errors in part (b)(ii) were 0.372^4 or $1 - 0.372^4$.

Q13.

The changes in values of p through this question proved to be quite challenging for some students. However, only the weakest students failed to score some marks in part (a)(i);

often due to the omission of $\begin{pmatrix} 30 \\ 2 \end{pmatrix}$ or an error in the indices. Part (a)(ii) caused more students a problem with a significant number trying to perform separate calculations for $p = 0.22$ and $p = 0.13$ before giving up since probabilities for neither value were tabulated. Those students who worked with $p = 0.35$ usually then made correct use of tables, rather than multiple formula calculations, although extracting a probability for an incorrect value (9 instead of 10) was still too prevalent. Disappointingly, given its regular examination on previous papers, for these and other students this uncertainty continued into part (a)(iii) which often resulted in the loss of at least one mark. Almost all students obtained correct values for the mean and variance in part (b)(i) although, for some students, there was some uncertainty as to whether the value of 17.16 was the variance or the standard deviation. Answers to part (b)(ii) were very disappointing with few students scoring more than one mark. All too often students compared the two means and the two standard deviations (using $\sqrt{17.6}$) but then did not even comment on the pairwise similarities or, more often than not, they simply stated 'results valid' or 'accept both claims'. In order to improve responses to such requests, teachers may wish to view the mark scheme to see what was required for the 3 marks.

Q14.

This question on the binomial distribution was a good source of marks for many candidates, with the more able often scoring most of the 15 marks available. Save for a small minority of candidates who calculated $P(R = 5)$ using the formula, part (a)(i) was usually answered correctly from the appropriate table in the supplied booklet. Again in part (a)(ii), answers were usually correct with only a minority of candidates giving $P(R \leq 10)$, $P(R \leq 9)$ or $1 - P(R \leq 9)$ as their answers. Answers to part (a)(iii) were almost always correct with evaluation by formula by far the more common method. As in previous series, part (a)(iv) caused candidates more difficulty, with many uncertain as to how to find $P(5 \leq R \leq 10)$. Whilst most candidates did attempt to subtract two cumulative probabilities, they often selected one, or even two, incorrect values. In part (b)(i), most candidates showed correctly that either $0.85 + (0.15 \times 0.80)$ or $1 - (0.15 \times 0.20)$ resulted in 0.97. Incorrect reasoning that also led to the given answer often assumed that a second shot was always made.

Although many correct answers were seen to part (b)(ii), the logic involved proved too much of a challenge for many candidates. Whilst many identified the correct model of B(50,0.97), most then attempted $P(S = 48)$ by formula. Most correct answers appeared to be obtained directly from a calculator's inbuilt cumulative binomial function rather than by $P(S' \leq 2)$ using tables for B(50,0.03). In answering part (b)(iii), most candidates attempted to use np but then failed to identify the correct values of n and p . As a result, $80 \times 0.80 = 64$ and $(80 \times 0.15) \times 0.80 = 9.6$ were common incorrect answers.

Q15.

This question produced a good range of marks, with most candidates scoring some marks in all parts. By far the most common and rewarding approach was to use tables as was intended. Candidates showed a good understanding of what was required, although there was the usual confusion of interpreting the requests and tables in terms of $<$, $\leq$, $>$ and $\geq$. This was particularly evident in part (a)(ii), where the common incorrect answer was $1 - 0.8202$, and in part (a)(iii), where $[0.9986 - 0.5443]$ or $[0.9901 - (1 - 0.5443)]$ were often seen. However, such errors were much less in evidence in part (b), with many candidates scoring most, and often all, of the 5 marks available. In part (c), most candidates used $n = 50$ instead of $n = 500$. Such candidates often scored 1 mark for the two follow-through answers of 7.5 and 6.375.

Q16.

Candidates scored well on this question, with many scoring at least 9 of the 11 marks. Answers to part (a)(i) were almost invariably correct, with very few candidates stopping at

$z = 1.25$ or finding $P(Z > 1.25)$. It was gratifying to see the absence of the use of $\sqrt{0.04}$ or 0.04^2 for the standard deviation. Few candidates acted on the symmetry of the request in part (a)(ii), but instead carried out the standardising of 57.1 before subtracting two areas. However, such working was usually clear and accurate with a correct final answer. Questions similar to part (b) caused a lot of confusion on previous papers. Here confusion was much less in evidence, with a significant proportion of candidates evaluating $[(a)(i)]^{16}$ in part (b). However, perhaps on the basis of a previous paper, $[(a)(i)]^6$ was not too rare

and standardisations of the form $\frac{57.2 - 57.15}{0.04}$ or $\frac{57.2 - 57.15}{0.04/\sqrt{16}}$ were still too prevalent.

Q17.

This question produced a good range of marks with most candidates scoring some marks in all parts. By far the most common and rewarding approach was to use tables as was intended; uses of formulae and inbuilt calculator functions were, in the main, much less successful. Candidates showed a good understanding of what was required, although there was the usual confusion of interpreting the requests and tables in terms of $<$, $\leq$, $>$ and $\geq$. This was particularly evident in part (a)(ii) where the common incorrect answer was $1 - 0.8202$ and in part (a)(iii) where $[0.9986 - 0.5443]$ or $[0.9901 - (1 - 0.5443)]$ were often seen. In a small but significant proportion of cases, carelessly misreading 0.5443 as 0.5543 lost marks. However, such errors were much less in evidence in part (b), with many candidates scoring most, and often all, of the 5 marks available.

Q18.

This was probably the best answered question on the paper with the better candidates often scoring the full 10 marks available. Apart from the small minority of candidates who found, by formulae, $P(R = 7)$ and $P(R = 10)$ in parts (a)(i) and (a)(ii) respectively, almost all candidates attempted to use the cumulative binomial tables in these parts. In part (a)(i), $P(R = 7)$ was usually found correctly but, in part (a)(ii), some candidates used $P(R \leq 11)$, instead of $P(R \leq 10)$, whilst others forgot to subtract their values obtained from the tables from 1. Part (a)(iii) caused some candidates more difficulty, with many uncertain as to how to determine $P(5 < R < 10)$. Whilst most candidates subtracted two cumulative probabilities, one or both were often incorrect values and, given that this type of question

has appeared regularly on previous papers, calculations such as $P(R \leq 9) - (1 - (P \leq 5))$ were particularly disappointing. Most answers to part (b), based on using the binomial formula, were correct.

Q19.

Almost all candidates knew how to standardise, and so the overwhelming majority scored the 3 marks in part (a)(i); the remainder usually scoring 2 marks for finding $P(X > 10.5)$. In part (a)(ii), again most candidates realised that a difference in areas was required, but calculating $P(X < 10.5) - P(X > 10.0)$ was all too common, as was

the evaluation of $P\left(Z < \frac{4}{3}\right)$ as $P(Z < 1.3)$ rather than $P(Z < 1.33)$.

Answers to part (b) almost invariably involved attempts at $P(\bar{X} > 10)$ instead of $(P(X > 10))^6$, this despite the fact that there was no mention of ‘mean’. Consequently, the majority of candidates scored 0 marks.

Q20.

This was probably the best answered question on the paper with the better candidates often scoring the full 14 marks available. Apart from the small minority of candidates who found, by formulae, $P(R = 7)$ and $P(R = 10)$ in parts (a)(i) and (a)(ii) respectively, almost all candidates attempted to use the cumulative binomial tables in these parts. In part (a)(i), $P(R = 7)$ was usually found correctly but, in part (a)(ii), some candidates used $P(R \leq 11)$, instead of $P(R \leq 10)$, whilst others forgot to subtract their values obtained from the tables from 1. Part (a)(iii) caused some candidates more difficulty, with many uncertain as to how to determine $P(5 < R < 10)$. Whilst most candidates subtracted two cumulative probabilities, one or both were often incorrect values and, given that this type of question has appeared regularly on previous papers, calculations such as $P(R \leq - (1 - P(R \leq 5)))$ were particularly disappointing.

Most answers to part (b), based on using the binomial formula, were correct, although a small minority of candidates used $n = 14$ or 42 or even $n = 20$, presumably so that tables could be used. Determination of the mean and variance in part (c)(i) was carried out correctly by almost all candidates. However, many candidates struggled to express their comparisons of results using appropriate language. Phrases involving “correct”, “incorrect” or “exact” were not appropriate, and neither was the word “different” when comparing 1.43 to 1.50; rather the two values were “similar” or “about/nearly the same”.

Q21.

Most candidates got off to a good start on the paper by scoring high marks on this question. Almost without exception, answers to part (a) were correct. In part (b), weaker candidates were uncertain as to which two probabilities to subtract, with $P(X \leq 14)$ and/or $P(X \leq 10)$ used instead of $P(X \leq 15)$ and $P(X \leq 9)$. Most candidates answered part (c) correctly, usually by using the appropriate formula rather than tables.

Q22.

Most candidates achieved 3 or 5 marks in answering part (a). Almost all candidates knew

how to standardise (without introducing an unnecessary continuity correction) and so the majority completed part (a)(i) accurately, but some then failed to apply the necessary area change correctly in part (a)(ii).

Candidates often scored either almost full marks or very few marks in part (b). Despite similar questions on previous papers, far too many candidates continued to perform a variety of standardisations and subtractions of areas for the 1 mark available in part (b)(i).

Few candidates answered part (b)(ii) correctly, despite a very similar request on the January 2010 paper. Some candidates with incorrect answers scored 1 mark for recognising the need to raise a probability to the power 6.

Q23.

This question on the binomial distribution was a good source of marks for many candidates, with the more able often scoring all the 14 marks available. Part (a)(i) was usually correct and the answer found from the appropriate table in the supplied booklet. A few candidates subtracted their correct answer from unity (giving $P(M > 10)$) whilst a few others calculated $P(M = 10)$ using the formula. Similar errors were seen in part (a)(ii), with $P(M \leq 5)$ and $P(M > 5)$ being the most common.

Part (a)(iii) caused candidates more difficulty, with many being uncertain as to how to find $P(6 < M < 12)$. Although most candidates did attempt to subtract two cumulative probabilities, they often selected incorrect values or sometimes even calculated $P(M \leq 11) - (1 - P(M \leq 6))$. The latter method was disappointing since this type of question has appeared regularly in the past. The vast majority of candidates used the binomial formula accurately in part (b).

Many candidates also scored full marks in part (c), whilst others only dealt with the male population. Most methods involved first finding the numbers of males and females, and then the respective number of left-handed of each gender, rather than treating it as a probability question. Thus it was quite well answered by even weak candidates.

Q24.

Many candidates scored at least 7 of the 8 marks available for this question by making use of the binomial formulae for a probability, mean and variance and for using cumulative binomial tables. Most candidates used the formula, rather than tables, to answer part (a)(i) and invariably scored full marks. In part (a)(ii), it was not unusual to see $P(X \leq 5)$ or even $P(X \leq 7)$ subtracted from $P(X \leq 12)$.

Answers to part (b) were usually correct.

Q25.

Most candidates scored at least 8 of the 10 marks available in part (a) by making appropriate use of the binomial formulae for a probability, mean and variance and for using cumulative binomial tables. Where marks were lost, it was usually for finding $P(X \leq 15)$ in part (a)(ii) and/or for subtracting $P(X \leq 15)$ from $P(X \leq 20)$ in part (a)(iii).

Only a very small number of candidates were able to give a satisfactory reason, such as the likely non-independence of senior citizens' travel, in part (b)(i). However most candidates gained the mark available in part (b)(ii), where a variety of acceptable reasons

were offered.

Q26.

Virtually all candidates used the appropriate binomial tables but frequently made one or more errors by finding $P(R \leq 10)$ in part (a)(i) and/or using $P(R \leq 9)$ or $P(R \leq 5)$ in part (a)(ii).

Candidates appeared well prepared in the use of the binomial formula, with almost all scoring full marks in part (b)(i). However, answers to part (b)(ii) were very disappointing. All too often, candidates calculated $P(S = 1)$ or $1 - P(S = 1)$ rather than $P(S \geq 1) = 1 - P(S = 0)$, with some even using $n = 22$ rather than 35.

In answering part (b)(iii), most candidates took note of the word 'correctly' and the number '120' in the question to then score both marks. However, a minority lost marks for stating

answers of 7.2 or $2.60(\sqrt{6.768})$. Part (b)(iv) proved beyond all but the high achievers. Whilst some candidates based their comments on the perceived practical difficulties in sorting letters, others decided that there was only one assumption and so made their comments unclear. Those candidates who did treat the statements as two separate assumptions, still often demonstrated little understanding of the implications of 'equal means' but 'different variances'. Many such candidates used the means to comment on independence and the variances to comment on the probability. As a result, the awarding of full marks for this final part was very rare.

Q27.

In part (a), most candidates realised that $B(50, 0.08)$ probabilities were required and so used the booklet provided. However, a significant proportion evaluated $P(X \geq x)$ as $1 - P(X \leq x)$ rather than $1 - P(X \leq x - 1)$ and so obtained incorrect answers. In part (b)(i), many candidates obtained correctly 0.684 from $(0.975)^{15}$ but then, for some unknown reason, some attempted part (b)(ii) as $1 - P(X \leq 1)$ by interpolation from $B(15, 0.02)$ and $B(15, 0.03)$, when all that was needed was $1 - 0.684 = 0.316$.

In part (c), many candidates recognised the need for $\{(a)(ii) \times (b)(i)\}$ but not for $\{(a)(i) \times (b)(ii)\}$. However, it was most pleasing to see that a significant number of candidates were able to score full marks on this question.

Q28.

Parts of this question proved a good source of marks for many candidates with the most able scoring all 12 marks. Answers to part (a) were usually correct and found, as was intended, from tables. A minority of candidates calculated $P(M = 15)$ using the formula. Most candidates used the correct formula in part (b), but a small minority approximated 0.29 by 0.30 so as to use tables, resulting in a loss of all of the 3 marks available. Answers to part (c)(i) showed an improvement in the knowledge of the relevant formulae. Only a very small proportion of candidates apparently chose to ignore the emboldened word '**do**' and used $p = 0.29$ rather than $p = 0.71$, resulting in a loss of at least 4 marks since the comment marks in part (c)(ii) were dependent upon correct answers in part (c)(i). Answers to part (c)(ii) were somewhat disappointing. Many candidates made no reference to their (correct) answers in part (c)(i) but simply stated that "the samples could not be random as they only included women". Only better candidates were able to score at least 2 of the 3 marks available. The mark lost was usually for stating that "since the samples were not random, the claim was not justified".

Q29.

Parts of this question proved a good source of marks for many candidates with the more able scoring all 15 marks. Answers to part (a)(i) were usually correct and found, as was intended, from tables. A minority of candidates calculated $P(M = 15)$ using the formula. As always, part (a)(ii) caused difficulties to many candidates. Whilst almost all of them attempted a subtraction using values from tables, many considered at least one incorrect value of M . The requirement of 'more than 10 but fewer than 20' translated to $P(10 < M < 20) = P(M \leq 19) - P(M \leq 10)$. Most candidates used the correct formula in part (b), but a significant minority approximated 0.29 by 0.30 so as to use tables, resulting in a loss of all of the 3 marks available. Answers to part (c)(i) showed an improvement in the knowledge of the relevant formulae. A small proportion of candidates apparently chose to ignore the emboldened word '**do**' and used $p = 0.29$ rather than $p = 0.71$, resulting in a loss of at least 4 marks since the comment marks in part (c)(ii) were dependent upon correct answers in part (c)(i). Nevertheless, many candidates were able to score at least 2 of the final 3 marks. When a mark was lost, it was usually for stating that "since the samples were not random, the claim was not justified". A number of candidates made no references to their answers in part (c)(i), but simply stated that "the samples could not be random as they only included women".

Q30.

Most candidates drew adequately a correct tree diagram but very few multiplied the probabilities (eg 0.25×0.30) which of course should, in total, add to one; there was no penalty for this 'omission'. However, almost all candidates did the necessary multiplications when answering part (b) correctly and so scored all 4 marks available. Answers to part (c) were much less impressive. Whilst the majority of candidates recognised the need for a binomial model with $n = 10$, most then used $p = \{(b)(i)\}$, 0.15 or 0.55, rather than $p = \{(b)(ii)\}$, presumably through not recognising the conditional nature of the request.

Q31.

Some candidates appeared to have no knowledge whatsoever of the binomial distribution. Of those candidates who did, many scored at least 6 marks. Almost all of them showed a correct expression in part (a) and then evaluated it correctly. Answers to part (b) often involved the usual confusion between ' $\leq$ ' and '<' when using Table 1 but the majority of candidates were able to score at least 1 mark. In part (c), too many candidates either misread the request, or were under the impression that $np(1 - p)$ gave the standard deviation.

Q32.

Many candidates scored at least 10 marks on this question. Almost all of them showed a correct expression in part (a) and then evaluated it correctly. Answers to part (b)(i) were also almost always correct with just a small minority of candidates evaluating $P(\text{Continental} = 12)$ from the formula. In part (b)(ii), there was the usual confusion between ' $\leq$ ' and '<' when using Table 1 but the majority of candidates were able to score at least 2 marks. In part (c), a minority of candidates did not use $p = 0.7$, whilst many more either misread the request or were under the impression that $np(1 - p)$ gave the standard deviation.

Q33.

Most candidates were able to make a worthwhile attempt at this question with many scoring full marks and only the weakest not recognising that binomial distributions were required. By far the most common approach in part (a) was use of the formula, rather than tables, usually with complete success. Thankfully however, the vast majority of candidates then switched to the use of Table 1 in part (b), although a small minority used their calculators. inbuilt binomial cumulative distribution function.

In part (b)(i), the use of $P(R \leq 15)$ for 'fewer than 15' was not uncommon but most candidates correctly equated 'more than 10' to $1 - P(R \leq 10)$, although perhaps by luck at times. In part (b)(iii), 'at least 12 but at most 18' was often incorrectly determined from $P(R \leq 18) - P(R \leq 12)$.

Q34.

This proved to be a good source of marks for many candidates with the more able often scoring full marks. Answers to part (a)(i) were usually correct from using Table 1 in the booklet provided. The minority using the formula often calculated $P(T = 3)$. Part (a)(ii) caused some difficulties. Whilst almost all candidates attempted a subtraction of values from Table 1, many had difficulty in translating 'on more than 10 days but fewer than 20 days' to $P(R \leq 19) - P(R \leq 10)$ with most errors involving the use of $P(R \leq 20)$ and/or $P(R \leq 9)$.

Knowledge of the relevant formulae needed to answer part (b)(i) was much improved. Failure to take the square root of 1.2 was the usual reason for forfeiting marks. As expected, part (b)(ii) was answered correctly by the vast majority of candidates. Answers to part (b)(iii) frequently contained comments that were valid, consistent and appropriate.

Q35.

Most candidates had a sound knowledge of calculating binomial probabilities. Many fewer candidates appeared to know how to calculate the mean and variance of a binomial distribution, whilst some appeared not to have studied the binomial distribution. A large majority of candidates scored full marks in parts (a)(i) & (ii). In part (a)(iii), confusion often arose as to whether the tabulated values required were for 7 or 8 and for 4 or 3. As a result many candidates lost some marks. A small minority of candidates opted for using the binomial formula sometimes with complete success. In part (b), those candidates with the necessary knowledge scored both marks. Others sometimes obtained the mean value by intuition but of course could make no headway with finding the variance. Yet again, far too many candidates appear unaware of the information provided in the first table on page 11 of the supplied booklet.